

Permutation-Invariant Set Encoders versus GRU Encoders for Active Diagnostic Probe Acquisition in Multimodal LLM Evaluation

Executive summary

This report studies one precise design choice for an ICLR-facing project on active multimodal reliability diagnosis: whether the acquired probe history should be encoded as an order-sensitive sequence with a gated recurrent unit, or as an unordered set with architectures such as Deep Sets, Set Transformer, or a masked slot-wise multilayer perceptron. The short answer is that if each probe is applied independently to the original image-question pair, and the diagnostic state should depend on *which* probes have been acquired rather than *the order* in which they arrived, then a permutation-invariant encoder is the more principled default. Deep Sets gives the cleanest inductive bias for this setting, while Set Transformer is the natural upgrade when cross-probe interactions matter. A GRU remains a strong engineering baseline and a good pilot implementation, although it is not the best final methodological centre for the paper. ¹

The literature already supplies all the neighbouring pieces. Deep Sets and Set Transformer cover invariant set modelling. Active feature acquisition covers cost-aware instance-wise information collection. Adaptive test-time compute work covers budgeted test-time policies and simple regret-style analyses. Recent multimodal hallucination benchmarks and detectors cover the evaluation surface on which such a method should be tested. The nearest public neighbour I found is a June 2026 paper on budgeted conformal evidence acquisition for VLM reliability, which argues that extra evidence can be better than abstention and shows that calibration must be performed on the full post-acquisition pipeline. However, that line is still about *verification and abstention*, not *source-level diagnostic measurement with set-valued behavioural evidence*. In the primary-source set reviewed here, I did **not** find a paper that jointly studies permutation-invariant encoding of acquired probe outcomes, active budgeted probe acquisition, and multimodal source diagnosis. That gap is where the novelty lies. ²

For your project, the strongest paper framing is not “we swapped a GRU for Deep Sets”. That is too small. The stronger claim is: **active multimodal diagnostic measurement should encode acquired probe outcomes as an unordered set of behavioural measurements, and this change improves stability, calibration, and generalisation under model and dataset shift**. If supported well, that is a real methodological contribution rather than a mild architecture ablation. ³

Novelty assessment

The core conceptual issue is very simple. A GRU assumes that the input is a meaningful sequence. That is appropriate when order itself carries semantics, as in language or time series. Here, each probe is usually a counterfactual measurement applied independently to the same original example: blank-image, blur, crop, relation swap, grounding, and similar probes do not form a natural temporal chain. If the final diagnosis should be the same whenever the same subset of probes has been observed, then order sensitivity is not a helpful bias; it is a possible source of spurious variation. Deep Sets was proposed exactly for functions on sets that should be invariant to permutation, and Set Transformer

extends this idea to interaction-aware set modelling while preserving the same invariance requirement.

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That does **not** mean a GRU is a bad baseline. In fact, it is a very good early model. GRUs are lightweight, easy to train, and commonly effective on short sequential inputs. Since your acquisition horizon is small, a GRU will likely work competitively in-distribution. The problem is not that it will fail outright. The problem is that a strict reviewer can ask a very uncomfortable question: *why use a sequence model when the measurement object is naturally a set?* If the paper has no good answer, the GRU becomes an avoidable conceptual weakness.

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The novelty question therefore becomes specific. Using Deep Sets or Set Transformer in isolation is not novel; those architectures are well established. Active feature acquisition is also not novel by itself. Budgeted test-time compute is likewise an active area. Multimodal hallucination diagnostics are now crowded as well. The novelty appears when these pieces are combined into one coherent problem: *instance-wise active acquisition of behavioural probes for multimodal diagnosis, with a state representation that respects the unordered nature of acquired evidence*. That combination is much more defensible than either “another set encoder paper” or “another active hallucination detector”.

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A second novelty lever is evaluation. Recent multimodal reliability work often studies single-pass internal-state detection, uncertainty scoring, selective abstention, or targeted verification. Those are important neighbours. Your project becomes more distinctive if it asks a different question: *how should a system choose which behavioural probes to acquire under a budget, and what representation best summarises the resulting evidence state?* That is a different research object from pre-generation detection, from score thresholding, and from post-hoc abstention.

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Literature review

Set encoders and sequence encoders

Deep Sets is the canonical starting point. Its main result characterises a broad family of permutation-invariant functions and motivates the familiar pooled form $\rho(\sum_i \phi(x_i))$. For your problem, that means the probe-state encoder can be built from per-probe evidence embeddings followed by symmetric pooling and a downstream predictor. Set Transformer keeps permutation invariance while allowing attention-based modelling of interactions among the set elements. Its induced-attention construction is especially relevant because your evidence sets are small enough that attention is affordable, although the model is still heavier than Deep Sets.

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GRU-style encoders come from the recurrent encoder-decoder line and remain useful because they are cheap, stable, and easy to update online. That practical advantage is real. However, their inductive bias is sequential rather than symmetric. In your setting, that means a GRU may learn order-specific hidden-state artefacts that have no behavioural meaning. From an ICLR-review perspective, this is exactly the sort of mismatch that reviewers like to press on.

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Active feature acquisition and value-of-information policies

The active feature acquisition literature is the closest formal neighbour. EDDI treats acquisition as Bayesian experimental design and scores candidate features by expected information gain on target variables. Reinforcement-learning-based work formulates acquisition as a sequential decision problem with measurement costs. More recent supervised latent-variable approaches try to avoid the instability

of reinforcement learning while still improving over purely myopic heuristics. A 2025 survey organises the field clearly and is very useful for paper positioning. ⁹

What transfers cleanly to your setting is the *budgeted, per-instance, value-of-information* viewpoint. What does **not** transfer automatically is the assumption that the “features” are ordinary tabular attributes. Your probes are behavioural counterfactual measurements with semantic structure, costs, and causal purpose. That difference is important for novelty framing. You are not merely choosing missing covariates. You are choosing which diagnostic interventions to run on a multimodal model. ¹⁰

Adaptive test-time compute and budgeted inference

The adaptive test-time compute literature is not the same problem, although it offers a useful analogy. Work such as Plan-and-Budget and constrained policy optimisation for reasoning LLMs asks which inputs deserve more test-time compute under an average or per-instance budget. Conformal Thinking studies how to control risk while stopping or continuing reasoning under a compute budget. These papers are helpful because they make budget allocation, policy imitation, and simple regret arguments familiar to reviewers. ¹¹

For your project, the strongest use of this literature is *not* to claim equivalence. The better move is to say that your work is the **measurement analogue** of adaptive test-time compute. Those methods spend more reasoning tokens. You spend more diagnostic measurements. The structural analogy is strong enough to be useful, while the scientific object remains distinct. ¹¹

Multimodal hallucination benchmarks and diagnostic methods

HallusionBench is still one of the most relevant public benchmarks because it explicitly targets entangled language hallucination and visual illusion rather than a single failure type. AMBER gives an LLM-free multidimensional evaluation over existence, attribute, and relation hallucination. IllusionBench makes visual illusion transfer a first-class held-out stress setting and includes trap cases that are particularly useful for exposing language-prior behaviour. PRE-HAL is important because it explicitly pushes beyond pure perception errors toward reasoning failures, especially relation reasoning. ¹²

On the detector side, HALP shows that hallucination risk can often be predicted before any token is generated using internal representations alone. VIB-Probe similarly studies probing internal heads and layers. VIHD is especially relevant because it uses *visual intervention* to calibrate semantic entropy in medical VQA, which is conceptually close to your probe-based evidence acquisition story even though it is domain-specific. The main implication is that the field already accepts targeted interventions and internal probing as legitimate reliability tools. What is missing is the active, budgeted, *set-encoded* behavioural acquisition story. ¹³

Calibration and evidence acquisition

This part of the literature matters a great deal for the final paper. Learning conformal abstention policies argues that adaptive policies can improve the trade-off among coverage, informativeness, and reliability. The closest neighbour is the June 2026 work on budgeted conformal evidence acquisition, which makes two findings that map directly onto your setting: first, extra evidence can be preferable to plain abstention; second, naive calibration can break once acquisition changes the score distribution, so the *entire post-acquisition pipeline* must be folded into calibration. That is highly relevant if you want calibrated diagnostic sets rather than just point predictions. ¹⁴

Key papers and models

The table below compares the most relevant papers and model reports in the reviewed primary-source set. The table is intentionally selective. It focuses on sources that matter directly for your novelty statement, baselines, and evaluation protocol. ¹⁵

Title	Year	Venue	Core idea	Relevance to your problem	Main strength	Main limitation
Deep Sets	2017	NeurIPS	Representation theorem and architecture for permutation-invariant set functions	Direct foundation for unordered probe-state encoding	Clean theoretical fit	Weak on interactions unless extended
Set Transformer	2019	ICML	Attention-based permutation-invariant set encoder	Direct foundation for interaction-aware set encoding	Models cross-probe interactions	More compute and tuning
Learning Phrase Representations using RNN Encoder-Decoder	2014	EMNLP	Sequence encoder-decoder with gated recurrence	Historical root of GRU-style baselines	Strong sequence baseline family	Sequential bias is mismatched here
Empirical Evaluation of Gated Recurrent Neural Networks on Sequence Modeling	2014	arXiv	Empirical GRU/LSTM comparison	Justifies GRU as a credible baseline	Practical efficiency	Not intended for unordered evidence
EDDI	2018	arXiv	Expected information gain for active feature acquisition	Strong template for value-of-information acquisition	Principled acquisition objective	Built for generic features, not behavioural probes
Reinforcement Learning with Efficient Active Feature Acquisition	2020	arXiv	RL-based cost-aware feature acquisition	Neighbour for acquisition policy learning	Sequential cost-aware control	Harder training and weaker interpretability

Title	Year	Venue	Core idea	Relevance to your problem	Main strength	Main limitation
A Survey on Active Feature Acquisition Strategies	2025	arXiv	Survey of cost-aware acquisition methods	Best framing source for related work	Clear taxonomy	Survey, not a method
Stochastic Encodings for Active Feature Acquisition	2025	arXiv	Supervised latent alternative to RL or greedy AFA	Suggests offline supervised acquisition learning	Strong modern acquisition baseline	Still feature-centric
Plan-and-Budget	2025	arXiv	Adaptive budget allocation for reasoning LLMs	Useful analogy for budget-conditioned control	Reviewer-friendly framing	Budget is extra reasoning, not extra measurement
Adaptive Test-Time Compute Allocation via Constrained Policy Optimisation	2026	arXiv	Solve-then-learn budget allocation with a clean reduction	Template for simple policy-regret claims	Strong imitation-style framing	Again about compute, not probes
Learning Conformal Abstention Policies	2025	arXiv	Adaptive conformal abstention for LLMs/VLMs	Calibration baseline for set-valued outputs	Strong risk-control framing	Abstention-oriented rather than diagnostic
Look Again Before You Abstain	2026	arXiv	Budgeted conformal evidence acquisition for VLM verification	Closest public neighbour	Evidence acquisition plus calibration	Focuses on verification, not source diagnosis
HallusionBench	2024	CVPR / arXiv	Diagnostic suite for entangled hallucination and illusion	Core benchmark for mixed failures	Human-crafted and difficult	Limited scale
AMBER	2023	arXiv	LLM-free existence/attribute/relation benchmark	Core evaluation benchmark	Cheap and multidimensional	Mostly benchmark-oriented

Title	Year	Venue	Core idea	Relevance to your problem	Main strength	Main limitation
IllusionBench	2025	arXiv	Large visual illusion benchmark with trap cases	Held-out illusion stress test	Strong for language-prior shift	Benchmark, not method
PRE-HAL with evidential conflict detector	2025	arXiv	Perception-versus-reasoning evaluation plus uncertainty detector	Strong held-out reasoning benchmark	Makes relation reasoning failures explicit	Not an active acquisition method
HALP	2026	EACL / arXiv	Pre-generation hallucination detection from internal states	Strong one-pass baseline	Very efficient	Lower diagnostic depth than active probing
VIB-Probe	2026	arXiv	Internal attention-head probing via information bottleneck	Strong probe-based detection baseline	White-box interpretability	Not active or budgeted
VIHD	2026	arXiv	Visual intervention-based hallucination detection in medical VQA	Neighbouring intervention method	Supports intervention-based detection story	Domain-specific
Qwen2.5-VL Technical Report	2025	arXiv	Modern open VLM report	Recommended training/eval model family	Strong visual grounding and localisation	Large engineering complexity
Molmo and PixMo	2024	arXiv	Fully open VLM with open data	Recommended model family	Strong reproducibility story	Platform-specific training stack
InternVL3	2025	arXiv	Strong open-source multimodal model	Recommended architecturally distinct model	Competitive capabilities	Large and complex training recipe
LLaVA-OneVision-1.5	2025	arXiv	Open multimodal framework with strong performance	Useful fourth-family evaluation option	Broad benchmark strength	Less central than Qwen/Molmo/InternVL for your story

Comparison of encoder families

The practical decision is not only about invariance. It is about the full trade-off among inductive bias, cost, interpretability, and generalisation. The table below states the comparison directly.

Dimension	Masked MLP	Deep Sets	Set Transformer	GRU
Permutation invariance	Yes, by canonical slots	Yes, by design	Yes, with invariant pooling	No
Sample efficiency	Often good under small data	Usually strongest default	Good if interactions matter	Good in-distribution, weaker if order artefacts appear
Compute cost	Lowest	Low	Moderate to high	Low
Interpretability	High	High	Moderate to high	Lower
Probe interaction modelling	Weak to moderate	Moderate	Strong	Strong, but sequential rather than symmetric
Order sensitivity	None	None	None	Intrinsic
Calibration stability	Potentially good	Potentially very good	Good if not overfit	More exposed to permutation drift
Shift generalisation	Good	Usually best bet	Good on hard mixed cases	Often more fragile
Best role in paper	Cheap invariant control	Main model	Interaction-aware ablation or upgrade	Strong baseline and pilot

This table points to the final recommendation. A masked multilayer perceptron is the right cheap invariant control. Deep Sets is the best *main* model. Set Transformer is the right interaction-aware upgrade. GRU should remain in the paper, although as a serious baseline rather than the conceptual centre. ¹⁶

Proposed architectural variants and exact evaluation protocol

Problem statement that is most defensible

The strongest problem statement is the following. A multimodal model is given an image-question pair and a set of possible diagnostic probes. Each probe is an extra behavioural measurement with a cost. The system must choose which probes to acquire under a budget and then output either a multi-label source signature or a calibrated set of plausible source signatures. The evidence state should be represented in a way that matches the semantics of acquisition. If probe order is not meaningful, then the encoder should be permutation invariant. This framing is much cleaner than a plain “predict the label cheaply” story because it ties the architecture directly to the measurement process. ¹⁷

Variant built around canonical slots

The first concrete variant is a **canonical masked slot multilayer perceptron**. Give each atomic probe its own slot. For each acquired probe, store an acquisition bit, a response-flip indicator, a confidence delta, a semantic-similarity score, a match or grounding score, and the probe cost. Concatenate all slots with clean-pass features, remaining budget, and optional model/dataset embeddings. Missing probes stay zeroed and are disambiguated by the acquisition bits.

The encoder is a masked multilayer perceptron. The heads are: one multi-label evidence head over the source bits, one optional simplified reporting head, one value-of-information head over unacquired actions, and one conformal score head for prediction-set construction. This variant is not novel as a neural block. Its novelty comes from the *behavioural evidence state design*, the *budgeted acquisition setting*, and the *diagnostic-set output*. It is the cheapest invariant baseline and therefore a very strong control. 18

Variant built around Deep Sets

The second and best default variant is a **Deep Sets evidence encoder**. Each acquired probe becomes an evidence token

$e_a = [\text{probe id embedding, flip, } \Delta c, \text{similarity, cost, probe family, clean-pass interaction features}]$.

If the acquired set at step t is S_t , encode it as

$$h_t = \rho \left(\sum_{a \in S_t} \phi(e_a) \right),$$

then append clean-pass summary features and remaining budget. The heads are the same as above: source-bit head, optional bit-pattern head, value-of-information head, and calibration head. Training uses multi-label loss for the source bits plus a regression or pairwise ranking loss for the value-of-information target. An additional permutation-consistency loss can be added even though the architecture is invariant, because it regularises implementation details and downstream heads. This is the variant most likely to satisfy a strict reviewer that the architecture matches the scientific object. 19

Variant built around Set Transformer

The third variant is a **budgeted Set Transformer**. The evidence tokens are the same. A small number of induced self-attention blocks is enough because the acquired evidence set is small. The set representation then feeds the same source, value-of-information, and calibration heads. This model should only become the preferred final version if probe interactions are demonstrably important. For example, the blank-image probe may mean something different when visual perturbations are stable than when they are unstable. That kind of interaction is exactly where Set Transformer can justify its extra cost. If those interaction gains are not there, keep Deep Sets as the main model and Set Transformer as an ablation. 20

Recommended datasets and model families

A strong protocol should separate *core training datasets* from *held-out stress datasets*. For core training and development, I recommend POPE or HOPE-style object hallucination data, HallusionBench, VizWiz-VQA, and a relation-heavy set such as VSR together with a GQA relation slice. For held-out evaluation,

IllusionBench and PRE-HAL are especially valuable because they stress illusion-heavy and reasoning-heavy shift regimes, while AMBER provides an independent multidimensional validation surface. ²¹

For model families, the cleanest public combination is Qwen2.5-VL as a strong modern anchor, Molmo as the fully open reproducible family, InternVL3 as an architecturally distinct high-performing family, and LLaVA-OneVision or a comparable LLaVA-family model as a fourth held-out family. If your earlier internal plan already uses Qwen3-VL or Gemma 4, I would treat them as practical substitutions, although the public primary-source base I reviewed most clearly supports Qwen2.5-VL, Molmo, InternVL3, and LLaVA-OneVision. ²²

Recommended baselines

The final paper should include at least the following baselines. First, clean-only scalar confidence and a clean-only multilayer perceptron predictor. Second, the GRU sequence encoder. Third, random acquisition and several fixed orders such as visual-first, blank-first, and relation-first. Fourth, uncertainty-greedy acquisition. Fifth, the full-probe teacher upper bound. Sixth, an oracle best-subset curve for each budget. Seventh, a strong one-pass internal-state detector such as HALP if white-box access is available. Eighth, a BCEA-style answer/abstain/acquire baseline wherever claim-level verification is meaningful. Those baselines are enough to convince reviewers that gains are not coming from weak controls. ²³

Exact permutation-invariance test

This part should be written with unusual precision because it is one of the clearest scientific tests in the whole paper.

For each example, precompute the full teacher cache for all probes. For each subset size $k \in \{1, 2, 3, 4\}$, either enumerate all probe subsets of size k if the atomic probe space is small or sample a fixed number of subsets per example. For each subset S , sample $P = 10$ random permutations $\pi_1(S), \dots, \pi_P(S)$. Run each encoder on every permutation and record: the predicted source probabilities p_π , the predicted diagnostic set Γ_π , the chosen next action a_π , and the hidden state h_π . Then report the mean pairwise Jensen-Shannon divergence among the p_π , the Jaccard disagreement among the Γ_π , the next-action agreement rate, the mean cosine distance or ℓ_2 distance among the hidden states, and any empirical calibration drift under permutations. A genuine set encoder should have near-zero output drift. A GRU usually will not unless heavily regularised. ¹⁶

During training, use three conditions: no permutation augmentation, random permutation augmentation only, and explicit consistency regularisation. A useful loss is

$$\mathcal{L}_{\text{perm}} = \mathbb{E}_{\pi, \pi'} \left[D_{\text{KL}}(p_\pi \parallel p_{\pi'}) + \lambda_a \|q_\pi - q_{\pi'}\|_2^2 + \lambda_h \|g(h_\pi) - g(h_{\pi'})\|_2^2 \right],$$

where q_π is the vector of action values or action logits and g is a small projection head. This gives a direct, reviewer-readable test of whether order sensitivity is scientific signal or arbitrary artefact. ²⁴

Metrics that matter most

For diagnosis, the load-bearing metrics should be multi-label micro and macro F1 over the source bits, Hamming loss, AUROC for incorrect-versus-correct prediction, calibrated-set coverage, and mean set size. For acquisition, the central figure should be the cost-diagnostic frontier and its area-under-frontier summary. For transfer, use leave-one-model-out, leave-one-dataset-out, and held-out stress on

IllusionBench and PRE-HAL. For calibration under shift, report empirical coverage honestly rather than over-claiming finite-sample guarantees where exchangeability no longer holds. This mirrors the lesson already emphasised in conformal evidence acquisition work. ²⁵

Theoretical claims, expected empirical signatures, and reviewer-facing figures

Claims you can make cleanly

The first theoretical claim is architectural invariance. A canonical masked-slot encoder is invariant to acquisition order by construction, because the slot state is identical for every permutation of the same acquired probe set. Deep Sets is permutation invariant by theorem. Set Transformer with invariant pooling likewise preserves permutation invariance. This gives a crisp formal motivation for why these encoders fit the task better than a recurrent sequence model. ²⁴

The second claim is a simple one-step value-of-information regret bound. Suppose the true next-action value is $V(h, a)$ and the predictor $\widehat{V}(h, a)$ satisfies

$$|\widehat{V}(h, a) - V(h, a)| \leq \epsilon$$

for every legal action a in state h . If the policy chooses

$$\hat{a} = \arg \max_a \widehat{V}(h, a),$$

then the chosen action's one-step regret is at most 2ϵ :

$$V(h, a^*) - V(h, \hat{a}) \leq 2\epsilon.$$

This is not a grand theorem, and that is precisely why it is useful. It is honest, relevant, and easy for reviewers to trust. Its role is to reduce acquisition quality to action-value estimation accuracy, which aligns well with the solve-then-learn logic in modern budget-allocation work. ²⁶

The third claim is calibration-aware. If the entire acquisition-and-prediction pipeline is frozen *before* calibration, and the same pipeline is applied to calibration and test examples under exchangeability, then split-conformal calibration of the post-acquisition score preserves marginal validity. This is the exact structural lesson drawn by recent budgeted conformal evidence acquisition work. Your paper need not overstate it. It is enough to say that once acquisition is part of the inference pipeline, calibration must be performed on the *full post-acquisition score*, not on a pre-acquisition score that is later modified by adaptive probing. ²⁷

Expected empirical signatures that would satisfy a strict ICLR reviewer

For the **masked multilayer perceptron**, the convincing signature is cheap, stable strength. It should beat the GRU on permutation consistency, match or slightly beat it on low-data settings, and remain highly competitive on the cost-diagnostic frontier. If this happens, the reviewer sees that a very simple invariant control already exposes the conceptual weakness of sequence encoding.

For **Deep Sets**, the convincing signature is broader. It should have near-zero permutation drift, beat the GRU on leave-one-model-out evaluation, and produce smaller calibrated diagnostic sets at matched coverage. If it also approaches the full-probe teacher using materially fewer extra passes, then the paper has a strong central story: the right inductive bias for behavioural evidence states matters.

For **Set Transformer**, the convincing signature should be *concentrated rather than generic*. It should win clearly on mixed cases, trap-illusion cases, or relation-heavy reasoning failures, not merely by tiny pooled improvements. Otherwise reviewers may see it as unnecessary complexity.

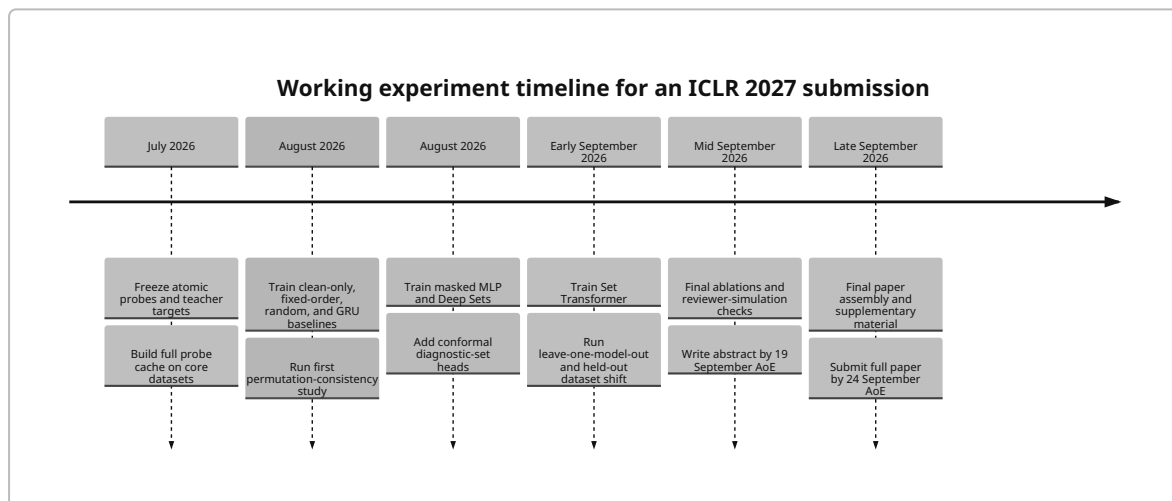
Across all models, the strongest overall result pattern is the following inequality:

clean-only < random/fixed < best learned policy < full teacher.

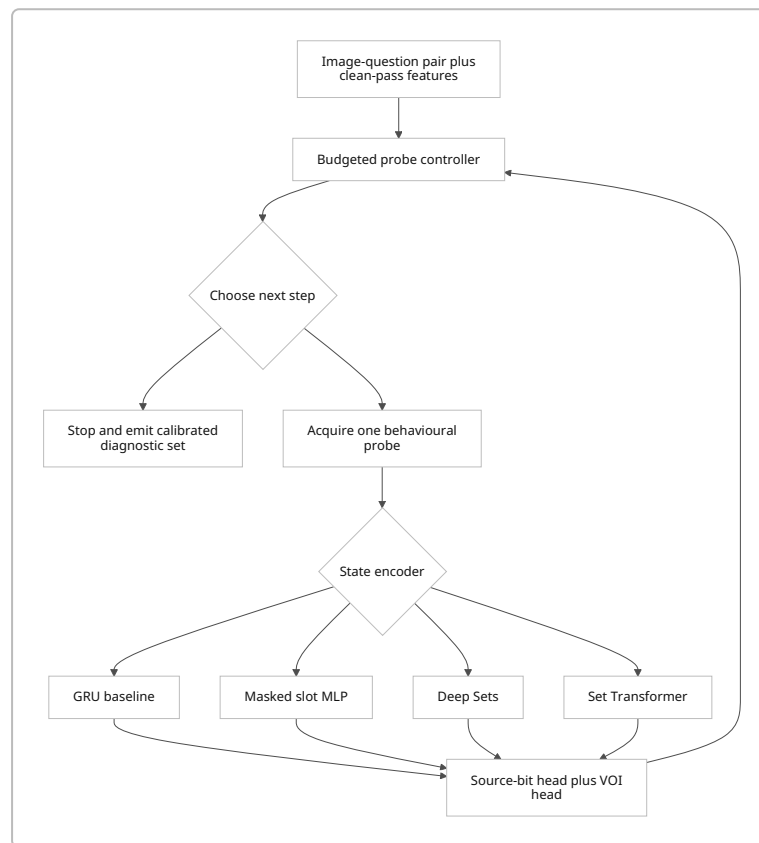
If the best learned policy is a set encoder and the GRU remains more order-sensitive and less stable under shift, the reviewer story becomes very clean. ²⁸

Working timeline aligned to the ICLR 2027 assumption

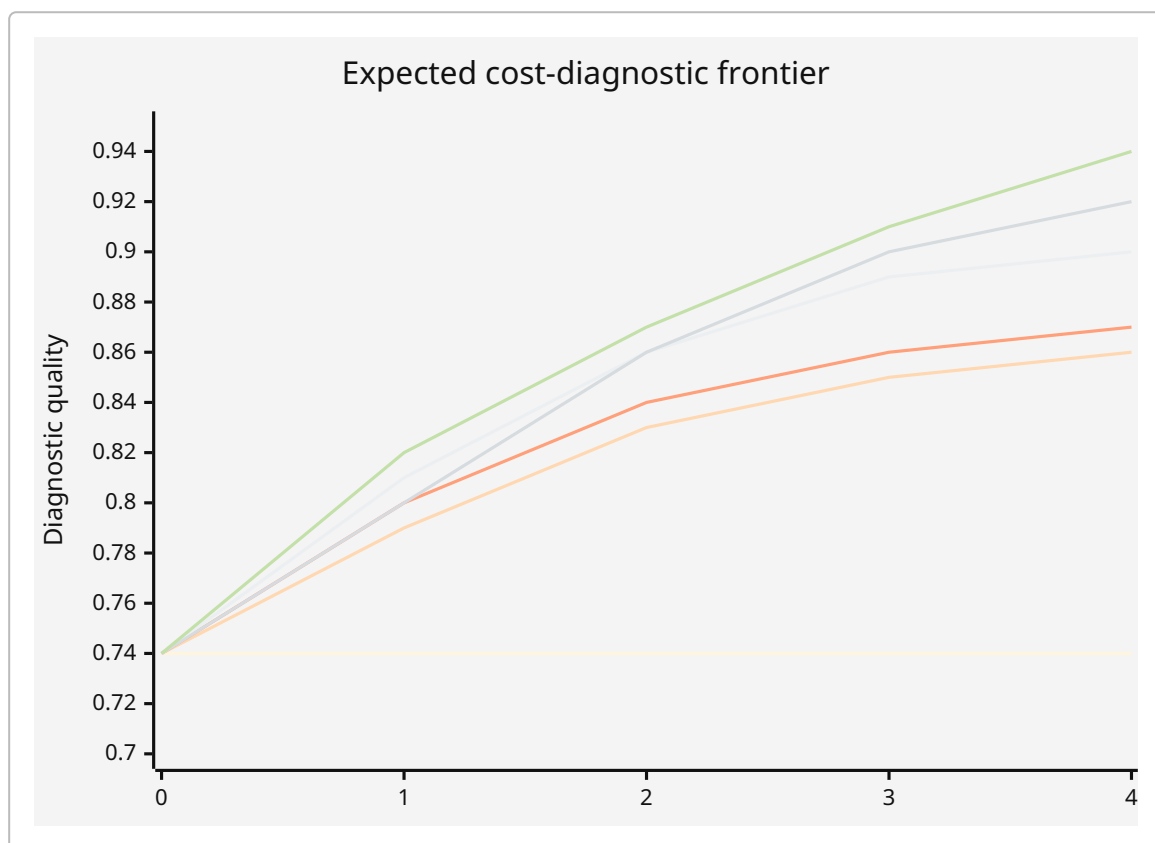
I adopt the often-used tentative ICLR schedule of abstract due on **19 September** and full paper due on **24 September** AoE, which matches the official ICLR 2026 submission schedule and is also the common working estimate used by current deadline trackers for ICLR 2027. ²⁹



Architecture-choice figure



Expected frontier figure



Final recommendation

My recommendation is straightforward. Keep the GRU in the paper as a serious baseline and possibly as the first pilot implementation. For the final ICLR paper, make **Deep Sets** the main encoder, keep the **masked multilayer perceptron** as the cheap invariant control, and use **Set Transformer** as the interaction-aware ablation. Frame the contribution around *active diagnostic measurement of unordered behavioural evidence*, not around an isolated encoder swap. If the invariant models show lower permutation drift, better sample efficiency, and stronger leave-one-model-out or held-out-dataset transfer while preserving or improving the cost-diagnostic frontier, then the architectural change is not cosmetic. It becomes the core methodological reason the work stands out. ³⁰

¹ ³ ⁴ ¹⁵ ¹⁶ ¹⁷ ¹⁹ ²⁴ ³⁰ Deep Sets

https://arxiv.org/abs/1703.06114?utm_source=chatgpt.com

² ⁶ ¹⁸ A Survey on Active Feature Acquisition Strategies

https://arxiv.org/abs/2502.11067?utm_source=chatgpt.com

⁵ ⁸ Learning Phrase Representations using RNN Encoder-Decoder for Statistical Machine Translation

https://arxiv.org/abs/1406.1078?utm_source=chatgpt.com

⁷ ¹³ ²³ HALP: Detecting Hallucinations in Vision-Language Models without Generating a Single Token

https://arxiv.org/abs/2603.05465?utm_source=chatgpt.com

⁹ ¹⁰ EDDI: Efficient Dynamic Discovery of High-Value Information with Partial VAE

https://arxiv.org/abs/1809.11142?utm_source=chatgpt.com

- 11 26 Adaptive Test-Time Compute Allocation for Reasoning LLMs via Constrained Policy Optimization
https://arxiv.org/abs/2604.14853?utm_source=chatgpt.com
- 12 HallusionBench: An Advanced Diagnostic Suite for Entangled Language Hallucination and Visual Illusion in Large Vision-Language Models
https://arxiv.org/abs/2310.14566?utm_source=chatgpt.com
- 14 25 Learning Conformal Abstention Policies for Adaptive Risk Management in Large Language and Vision-Language Models
https://arxiv.org/abs/2502.06884?utm_source=chatgpt.com
- 20 Set Transformer: A Framework for Attention-based Permutation-Invariant Neural Networks
https://arxiv.org/abs/1810.00825?utm_source=chatgpt.com
- 21 Evaluating Object Hallucination in Large Vision-Language Models
https://arxiv.org/abs/2305.10355?utm_source=chatgpt.com
- 22 Qwen2.5-VL Technical Report
https://arxiv.org/abs/2502.13923?utm_source=chatgpt.com
- 27 Look Again Before You Abstain: Budgeted Conformal Evidence Acquisition for Reliable Vision-Language Model
<https://arxiv.org/abs/2606.16667>
- 28 IllusionBench: A Large-scale and Comprehensive Benchmark for Visual Illusion Understanding in Vision-Language Models
https://arxiv.org/abs/2501.00848?utm_source=chatgpt.com
- 29 ICLR 2026 Call for Papers
https://iclr.cc/Conferences/2026/CallForPapers?utm_source=chatgpt.com